

Geometric time-dependent density-functional theory

**Éric Cancès¹, Théo Duez¹, Jari van Gog²,
Asbjørn B. Lauritsen³, Mathieu Lewin³, Julien Toulouse²**

¹ École des Ponts, Institut Polytechnique de Paris, Inria and CNRS, France

² Sorbonne Université and CNRS, France

³ Université Paris-Dauphine–PSL and CNRS, France

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1 Geometric TDDFT

2 Geometric TDKS

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A general formulation of TDDFT

- ▶ Consider the N -electron **Hamiltonian** $\hat{H}(t) := \hat{T} + \hat{V}_{\text{ee}} + \hat{V}_{\text{ext}}(t)$ and the **time-dependent Schrödinger equation (TDSE)**

$$\begin{cases} i\partial_t \Psi^S(t) = \hat{H}(t) \Psi^S(t) \\ \Psi^S(0) = \Psi_0 \end{cases} \text{ leading to the time-dependent density } \rho^S := \rho_{\Psi^S}$$

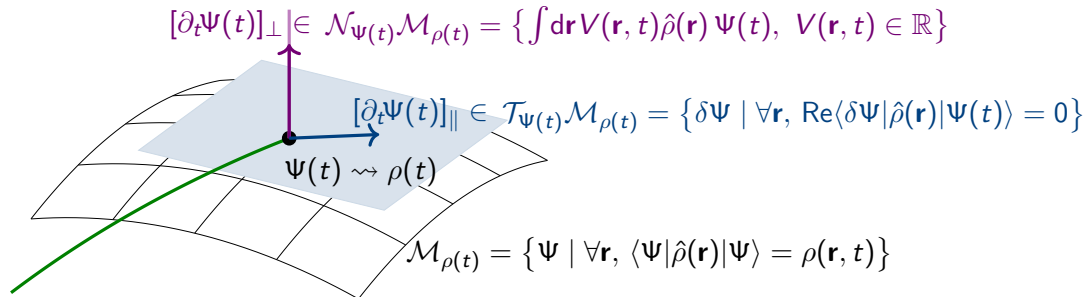
- ▶ We want to find the exact ρ^S from an equivalent equation involving the density only
- ▶ For given Ψ_0 , V_{ext} , ρ , find $\hat{F}(t)$ in a class of operators (e.g., 1-e local potentials) such that

$$\begin{cases} i\partial_t \Psi(t) = (\hat{H}(t) + \hat{F}(t)) \Psi(t) \\ \Psi(0) = \Psi_0 \end{cases} \text{ gives the prescribed density } \rho_{\Psi} = \rho$$

- ▶ If we have a **Runge–Gross theorem** for this class of operators, it gives us a **unique** way of defining the **functional** $\hat{F}[\Psi_0, V_{\text{ext}}, \rho](t)$
- ▶ We then have a **fundamental equation** for ρ^S : $\hat{F}[\Psi_0, V_{\text{ext}}, \rho] = 0 \iff \rho = \rho^S$

Geometric interpretation of the general formulation of TDDFT

- ▶ Choosing a $\hat{F}(t)$ amounts to **choosing the wave-function velocity** $\partial_t \Psi(t)$
- ▶ To reproduce ρ , there is **freedom** in choosing $\partial_t \Psi(t)$ which can be understood **geometrically**



- ▶ $\partial_t \Psi(t)$ can be decomposed into **tangent and normal components**

$$\partial_t \Psi(t) = [\partial_t \Psi(t)]_{\parallel} + [\partial_t \Psi(t)]_{\perp}$$

where $[\partial_t \Psi(t)]_{\parallel}$ is **completely free** and only $[\partial_t \Psi(t)]_{\perp}$ is **fixed** by $\partial_t \rho(t)$

Standard TDDFT (1/2)

- **Standard TDDFT** amounts to choosing $\hat{F}(t)$ using the **variational principle (VP)**

$$\text{stat}_{\Psi \rightsquigarrow \rho} \int_0^T dt \langle \Psi(t) | i\partial_t - \hat{H}(t) | \Psi(t) \rangle \iff \underbrace{\text{Re} \langle \delta\Psi | i\partial_t - \hat{H}(t) | \Psi(t) \rangle}_{\hat{F}(t)\Psi(t)} = 0, \forall \delta\Psi \in \mathcal{T}_{\Psi(t)}\mathcal{M}_{\rho(t)}$$

- This means that $\hat{F}(t)\Psi(t) \in \mathcal{N}_{\Psi(t)}\mathcal{M}_{\rho(t)}$, so $\hat{F}(t)$ is of the form

$$\hat{F}^{\text{VP}}(t) := \int d\mathbf{r} V(\mathbf{r}, t) \hat{\rho}(\mathbf{r}) \quad \text{with } V(\mathbf{r}, t) \in \mathbb{R}$$

and the **Runge–Gross theorem** (1984) gives us a **unique functional** $V = V[\Psi_0, V_{\text{ext}}, \rho]$

- We have a **fundamental equation** for the exact density ρ^S :

$$V[\Psi_0, V_{\text{ext}}, \rho]_{+C(t)} = 0 \iff \rho = \rho^S$$

- or, defining the **universal functional potential** $V[\Psi_0, \rho] := V[\Psi_0, 0, \rho]$,

$$V[\Psi_0, \rho]_{+C(t)} = V_{\text{ext}} \iff \rho = \rho^S$$

Standard TDDFT (2/2)

- ▶ Thus, in **standard TDDFT**, a prescribed density ρ is reproduced from an effective TDSE with a **(real-valued) local potential**

$$\begin{cases} i\partial_t \Psi(t) = \left(\hat{T} + \hat{V}_{ee} + \hat{V}(t) \right) \Psi(t) \\ \Psi(0) = \Psi_0 \end{cases} \quad \text{where } \hat{V}(t) := \int d\mathbf{r} V[\Psi_0, \rho](\mathbf{r}, t) \hat{\rho}(\mathbf{r})$$

- ▶ The **continuity equation** is always satisfied

$$\partial_t \rho(\mathbf{r}, t) + \nabla \cdot \mathbf{j}_{\Psi(t)}(\mathbf{r}) = 0$$

and does not explicitly involve the potential $V(\mathbf{r}, t)$

- ▶ $V(\mathbf{r}, t)$ can be found from **second-order time-derivative of the density** (van Leeuwen 1999)

$$\partial_t^2 \rho(\mathbf{r}, t) = -2 \int K_{\Psi(t)}(\mathbf{r}, \mathbf{r}') V(\mathbf{r}', t) d\mathbf{r}' + q_{\Psi(t)}(\mathbf{r})$$

with $q_{\Psi}(\mathbf{r}) = -\langle \Psi | [\hat{T} + \hat{V}_{ee}, [\hat{T}, \hat{\rho}(\mathbf{r})]] | \Psi \rangle$ and

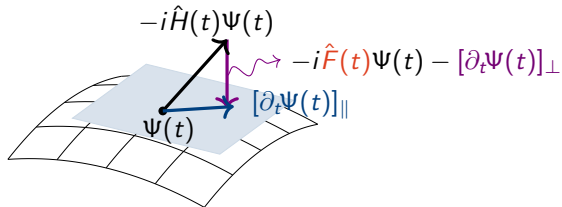
$$K_{\Psi}(\mathbf{r}, \mathbf{r}') := \frac{1}{2} \langle \Psi | [\hat{\rho}(\mathbf{r}'), [\hat{T}, \hat{\rho}(\mathbf{r})]] | \Psi \rangle = \frac{1}{2} \nabla_{\mathbf{r}} \cdot (\rho_{\Psi}(\mathbf{r}) \delta'(\mathbf{r}' - \mathbf{r}))$$

Geometric TDDFT (1/2)

- In our **geometric TDDFT**, we choose $\hat{F}(t)$ using the **geometric principle (GP)** (or McLachlan principle)

$$\min_{\partial_t \Psi(t) \rightsquigarrow \partial_t \rho(t)} \left\| \underbrace{i \partial_t \Psi(t) - \hat{H}(t) \Psi(t)}_{\hat{F}(t) \Psi(t)} \right\| \iff [\partial_t \Psi(t)]_{\parallel} = \hat{P}_{\mathcal{T}_{\Psi(t)} \mathcal{M}_{\rho(t)}} \left(-i \hat{H}(t) \Psi(t) \right)$$

i.e., we make the norm of $\hat{F}(t) \Psi(t)$ minimal
 $\iff [\partial_t \Psi(t)]_{\parallel}$ is the orthogonal projection of $-i \hat{H}(t) \Psi(t)$ onto the tangent space



- This leads to $-i \hat{F}(t) \Psi(t) \in \mathcal{N}_{\Psi(t)} \mathcal{M}_{\rho(t)}$, so $\hat{F}(t)$ is of the form

$$\hat{F}^{\text{GP}}(t) := i \int d\mathbf{r} W(\mathbf{r}, t) \hat{\rho}(\mathbf{r}) \quad \text{with } W(\mathbf{r}, t) \in \mathbb{R}$$

and our **geometric Runge–Gross theorem** gives a **unique functional** $W = W[\Psi_0, V_{\text{ext}}, \rho]$

- Our new **fundamental equation** for ρ^S is: $W[\Psi_0, V_{\text{ext}}, \rho] = 0 \iff \rho = \rho^S$

Geometric TDDFT (2/2)

- ▶ Thus, in **geometric TDDFT**, a prescribed density ρ is reproduced from an effective TDSE with a **imaginary local “potential”**

$$\begin{cases} i\partial_t \Psi(t) = \left(\hat{H}(t) + i\hat{W}(t) \right) \Psi(t) \\ \Psi(0) = \Psi_0 \end{cases} \quad \text{where } \hat{W}(t) := \int d\mathbf{r} W[\Psi_0, V_{\text{ext}}, \rho](\mathbf{r}, t) \hat{\rho}(\mathbf{r})$$

- ▶ $\hat{W}(t)$ reproduces $\rho(t)$, so it preserves the norm of $\Psi(t)$, and we have $\langle \Psi(t) | \hat{W}(t) | \Psi(t) \rangle = 0$
- ▶ This can be used to rewrite the effective TDSE with a **nonlocal Hermitian operator**

$$\begin{cases} i\partial_t \Psi(t) = \left(\hat{H}(t) + i \left[\hat{W}(t), |\Psi(t)\rangle \langle \Psi(t)| \right] \right) \Psi(t) \\ \Psi(0) = \Psi_0 \end{cases}$$

- ▶ The exact $W(\mathbf{r}, t)$ can be found from the **modified continuity equation**

$$\partial_t \rho(\mathbf{r}, t) + \nabla \cdot \mathbf{j}_{\Psi(t)}(\mathbf{r}) = 2 \int S_{\Psi(t)}(\mathbf{r}, \mathbf{r}') W(\mathbf{r}', t) d\mathbf{r}' \quad \text{with} \quad S_{\Psi}(\mathbf{r}, \mathbf{r}') := \langle \Psi | \hat{\rho}(\mathbf{r}) \hat{\rho}(\mathbf{r}') | \Psi \rangle$$

- ▶ $W(\mathbf{r}, t)$ generates **local sources and sinks of density**

1 Geometric TDDFT

2 Geometric TDKS

Geometric TDKS

- ▶ Setting $\hat{V}_{ee} = 0$, a prescribed density ρ is reproduced from an effective **non-interacting** TDSE

$$\begin{cases} i\partial_t \Phi(t) = \left(\hat{T} + \hat{V}_{\text{ext}}(t) + i\hat{W}_0(t) \right) \Phi(t) \\ \Phi(0) = \Phi_0 \end{cases} \quad \text{with } \hat{W}_0(t) := \int d\mathbf{r} W_0[\Phi_0, V_{\text{ext}}, \rho](\mathbf{r}, t) \hat{\rho}(\mathbf{r})$$

where the initial Slater determinant Φ_0 has density $\rho(\mathbf{r}, 0)$

- ▶ If we input $\rho = \rho^S$ then $\Phi(t)$ correctly reproduces the physical density ρ^S . However, this cannot be used to predict ρ^S since we do have any information about $\hat{V}_{ee}$ yet
- ▶ To obtain a predictive scheme, we define the **geometric KS (GKS) universal functional**

$$W^{\text{GKS}}[\Phi_0, \Psi_0, \rho] := W_0[\Phi_0, V[\Psi_0, \rho], \rho]$$

where $V[\Psi_0, \rho]$ is the **interacting universal functional potential** of standard TDDFT

- ▶ We thus obtain the **geometric TDKS scheme**

$$\begin{cases} i\partial_t \Phi(t) = \left(\hat{T} + \hat{V}_{\text{ext}}(t) + i\hat{W}^{\text{GKS}}[\Phi_0, \Psi_0, \rho_\Phi](t) \right) \Phi(t) \\ \Phi(0) = \Phi_0 \end{cases}$$

TDKS with geometric correction

- ▶ In practice, we may want to use a W only to correct an approximate Hxc potential $V_{\text{Hxc}}^{\text{app}}[\rho]$
- ▶ This gives us the **TDKS scheme with geometric correction**

$$\begin{cases} i\partial_t\Phi(t) = \left(\hat{T} + \hat{V}_{\text{ext}}(t) + \hat{V}_{\text{Hxc}}^{\text{app}}[\rho_\Phi](t) + i\hat{W}^{\text{corr}}[\Phi_0, \Psi_0, \rho_\Phi](t) \right) \Phi(t) \\ \Phi(0) = \Phi_0 \end{cases}$$

where $W^{\text{corr}}[\Phi_0, \Psi_0, \rho] := W_0[\Phi_0, V[\Psi_0, \rho] + V_{\text{Hxc}}^{\text{app}}[\rho], \rho]$

- ▶ In terms of **orbitals**, to keep them orthonormal, we use the **Hermitian commutator form**

$$i\partial_t\varphi_n(t) = \left(-\frac{\Delta}{2} + V_{\text{ext}}(t) + V_{\text{Hxc}}^{\text{app}}(t) + i[W^{\text{corr}}(t), \gamma(t)] \right) \varphi_n(t)$$

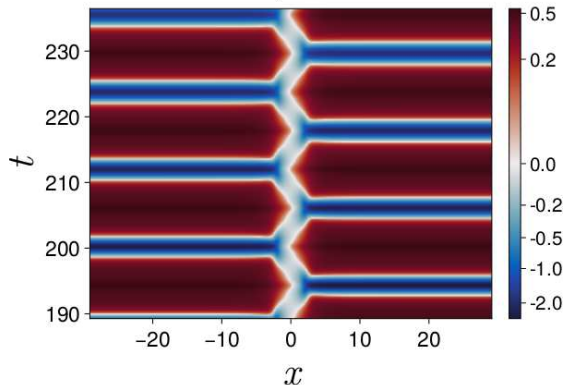
where $\gamma(t) := \sum_{n=1}^N |\varphi_n(t)\rangle\langle\varphi_n(t)|$ is the one-electron density matrix

Rabi oscillations in 1D Helium with soft-Coulomb potential

► Non-adiabatic contributions in:

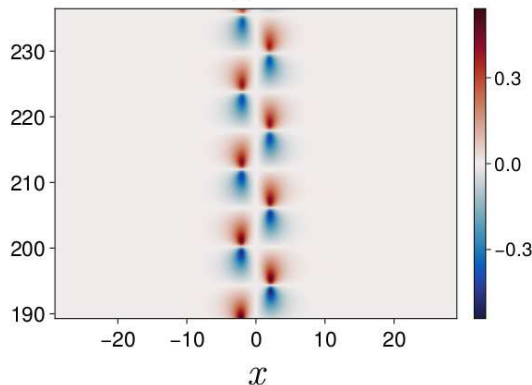
standard TDKS

V^{na}



TDKS with geometric correction

W^{na}



- In **standard TDKS**, $V^{\text{na}}(x, t)$ has **fast-evolving spatially-extended steps**
- In **TDKS with geometric correction**, $W^{\text{na}}(x, t)$ is **much more localized**